Venus

Overview

Venus is the second planet from the Sun and the closest planetary neighbour to Earth, orbiting at an average distance of 108 million km. With a diameter of 12,104 km, it is nearly the same size as Earth and is often called Earth's twin — but the resemblance is only superficial, as Venus is one of the most inhospitable environments in the Solar System. Its thick, toxic atmosphere creates surface conditions so extreme that early Soviet landers survived for only 23–127 minutes before being crushed and melted. Venus is the brightest natural object in the night sky after the Moon, reaching a magnitude of up to -4.9, because its dense cloud cover reflects about 70 % of incoming sunlight — the highest albedo of any planet. Despite its brightness and proximity, Venus was not well understood until the space age, when radar and spacecraft finally pierced its permanent cloud cover.

Atmosphere and Greenhouse Effect

Venus's atmosphere is composed of 96.5 % carbon dioxide and 3.5 % nitrogen, with trace amounts of sulfur dioxide, water vapour, and other gases. This composition creates a catastrophic runaway greenhouse effect: solar radiation penetrates the atmosphere and warms the surface, but the outgoing infrared radiation is almost entirely absorbed and re-emitted by the dense CO_2, trapping heat and raising the surface temperature to an average of 465 °C — hot enough to melt lead. Atmospheric pressure at the surface is 92 bar, about 92 times that of Earth at sea level, equivalent to the pressure experienced 900 metres underwater. The cloud layers, composed primarily of sulfuric acid droplets, extend from about 45 to 70 km altitude and reflect most incoming sunlight. Despite this reflection, the greenhouse warming is so intense that Venus receives only about a quarter of the solar energy reaching Earth's surface, yet is still far hotter than Mercury, which receives far more direct sunlight.

Surface and Geology

The surface of Venus, invisible from Earth due to permanent cloud cover, was revealed in detail by NASA's Magellan spacecraft, which mapped 98 % of the surface with radar between 1990 and 1994. About 80 % of the surface consists of volcanic plains — vast flat regions of solidified lava flows indicating extensive past (and possibly ongoing) volcanic activity. Two major highland regions, Ishtar Terra in the north (roughly the size of Australia) and Aphrodite Terra near the equator (roughly the size of South America), dominate the topography. Maxwell Montes, within Ishtar Terra, is the highest mountain on Venus at 11 km above the mean surface elevation — taller than Mount Everest above sea level. Venus has over 1,600 major volcanoes; a 2023 reanalysis of Magellan radar images confirmed a volcanic vent changed shape between 1990 and 1992, providing the first direct evidence of currently active volcanism on Venus. Interestingly, Venus has very few impact craters and none smaller than about 3 km, because the thick atmosphere burns up smaller incoming bodies before they can hit the surface.

Rotation and Orbit

Venus has one of the most unusual rotation patterns in the Solar System. It rotates in the retrograde direction — from east to west — opposite to most planets, meaning the Sun rises in the west and sets in the east on Venus. Its rotation is also extremely slow: a Venusian sidereal day (one full rotation relative to the stars) lasts 243 Earth days, which is longer than its orbital year of 225 Earth days — the only planet where the day is longer than the year. Because of the retrograde rotation, a solar day on Venus (sunrise to sunrise) is actually 117 Earth days, slightly shorter than the sidereal day. The leading hypothesis for Venus's retrograde rotation is that a massive collision early in its history flipped the planet upside down, or alternatively that gravitational resonances with Earth and solar tidal forces gradually slowed and reversed its original prograde spin over billions of years.

Exploration and Future Missions

Venus has been visited by more spacecraft than any other planet aside from Mars. The Soviet Venera programme achieved the greatest early success, landing multiple probes on the surface between 1970 and 1984, transmitting the first images from the Venusian surface and measuring its crushing pressure and extreme temperatures. NASA's Magellan mission (1990–1994) produced the most complete radar map of the surface and measured gravitational anomalies that revealed interior structure. ESA's Venus Express orbited Venus from 2006 to 2014, studying the atmosphere in detail and finding evidence of past liquid water and ongoing volcanism. A new wave of Venus missions is planned for the late 2020s and 2030s: NASA's DAVINCI will probe the atmosphere during descent; NASA's VERITAS will map the surface at higher resolution than Magellan; and ESA's EnVision will study Venus from crust to atmosphere to understand whether it was once habitable and why it diverged so dramatically from Earth.